

2025 IEEE Symposium on CI for Energy, Transport and Environmental Sustainability (IEEE CIETES)					
18th March 2025					
Session	Time	Room	#Papers	Title	Authors
oral S1-A	14:00 - 15:00	Cosmos 3a	6	Game-Theoretic Approach for Cooperative Planning of Active Distribution Networks and Private Electric Vehicle Charging Stations	Mejia , Mario; Macedo, Leonardo H.; Muñoz-Delgado, Gregorio; Contreras, Javier; Franco, John F.
long S1-A	15:30 - 16:30	Cosmos 1&2		On Scalable Design for User-Centric Multi-Modal Shared E-Mobility Systems using MILP and Modified Dijkstra's Algorithm	Shah, Maqsood ; Li, Ji; Liu, Mingming
				Split Computing in Autonomous Mobility for Efficient Semantic Segmentation using Transformers	Ulicny, Chris; Rauch, Robert*; Gazda, Juraj; Becvar, Zdenek
				Optimizing Takagi-Sugeno Fuzzy Models For Improving Leak Detection in Water Distribution Networks	Khaled, Qusai; Kaymak, Uzay; Genga, Laura
				Dynamic Ensemble Short-Term Load Forecasting for Residential Building with Low Data Availability: SARIMA vs. LSTM	Palaniswamy, Lakshimi Narayanan; Kappler, Tim; Munzke, Nina; Hiller, Marc
				Optimizing Industrial E-Waste Recycling with Attention-Driven Deep Learning for PCB Segmentation Using Hyperspectral Imaging	Trishna Barman; Sonya Coleman; Dermot Kerr; Justin Quinn; Shane Harrigan
short S1-A	16:30 - 17:30	Cosmos 1&2	6	Stabilise Power Grids from fluctuating Renewable Energy Sources production with Artificial Immune System	Antoine Lihard; Wei Pang
				Adverse Weather Benchmarks and Dataset for Object Detection in Autonomous Driving	Dominik Weikert; Adrian Köring; Christoph Steup
				Multi-Objective Optimization Algorithms For Energy Management System in Microgrids Including Control Strategy	Saiful Islam; Sanaz Mostaghim; Michael Hartmann
				Asynchronous differential evolution with Lissajous mutation for efficient energy management of plug-in electric vehicles	Arturo Valdivia; Angel Casas-Ordaz; Itzel Aranguren; Diego Oliva; Seyed Jalaeddin Mousavirad
				Data-Driven Adaptive Control for Frequency Perturbations in Power Systems with PV-SmartParks	Ali Arzani; Ganesh K. Venayagamoorthy
19th March 2025					
Session	Time	Room	#Papers	Title	Authors
short S1-B	12:00-13:00	Cosmos 1&2	5	Online Detection of Water Contamination Under Concept Drift	jin li; kleanthis Malialis; Stelios Vrachimis; Marios Polycarpou
				Urban Water Consumption Forecasting Using Deep Learning and Correlated District Metered Areas	Kleanthis Malialis; Nefeli Mavri; Stelios Vrachimis; Marios Kyriakou; Demetrios Eliades; Marios Polycarpou
				Multi-Objective Reinforcement Learning for the Control of Storm-water Systems under Distributional Shift	Daisy Welham; Sara Sharifzadeh; Liam Butler; Chedly Tizaoui
				Permutation Optimization using Multivariate Dependent Estimation of Distribution Algorithm	Yuyang Guo; Chu-ge Wu; Zhuo Li; Yuanqing Xia
				Timely predicting Power Mosfet failure in EV: AI based methodology and deployment at the edge	Robin Faro; Alessandro Strano; Francesco Cancelliere; Raffaele Mineo
LBP S1	14:00 - 15:00	Cosmos 1&2	6	Enhancing Vehicle Navigation in GNSS-Limited Environments with Georeferenced Snow Poles	Bavirisetti, Durga Prasad; Berget, Gunhild ; Tabassum, Shaira ; Kiss, Gabriel ; Arnesen, Petter ; Seter, Hanne ; Lindseth, Frank
				Deep Learning-based segmentation for Automated Produced Water Re-Injection Analysis	Saiti, Evdokia; Hatiboglu, Anil; Azizov, Ilgar; Ahmed, Husnain; Shardt, Nadia; Oye, Gisle
				Designing High-Performance and Scalable Solutions for Virtual Power Plants	Rodrigues, Rciardo; Abbasi, Ali; Alves, Filipe; Gomes, Jean; Ribeiro, Rui; Petiz, Maria; Sobral, Joao Luis
				Decentralised Traffic Incident Detection via Network Lasso	Zhu, Qiyuan; Qin, Kai; Dia, Hussein; Grzybowska, Hanna
				Optimizing Voltage Control in Residential Photovoltaic Inverters through Personalized Federated Learning: Experimental Insights	Kaneko, Nanae; Fujimoto, Yu; Takahashi, So; Kaneko, Akihisa; Yutaka, Ino; Hayashi, Yasuhiro
				A Digital Twin for Flexibility Management in Distribution Network	EL IALLI, Mohamad; Razi, Reza; Bruyere, Antoine; Francois, Bruno; Soares, Joao; Vale, Zita
oral S1-B	15:00-16:00	Cosmos 3c	5	Active Learning guided Gaussian Process Regressor for Non-Linear PDE Modeling for environmental simulations	Soumen Sinha; Lara Bastos; Rajen Kumar Sinha
long S1-B	16:00-17:00	Cosmos 1&2		Real-Time Lithium-Ion Battery Capacity Curve Recovery and High-Precision RUL Prediction in the Presence of Capacity Data Loss	Yixin Nie; Anqi Wang; Zhi Lu; Zhiyu Zhang; Liqin Yan; Jingying Xie; Fan Yang
				Comparative Analysis of Machine Learning-Based Imputation Techniques for Air Quality Datasets with High Missing Data Rates	Sen Yan; David O'Connor; Xiaojun Wang; Noel Connor; Alan Smeaton; Mingming Liu
				A Genetic Algorithm based Deterministic Multi-step Peer-to-Peer Power Routing for Energy Internet in Sparsely Connected Microgrid Communities	Neethu Maya; Narasimman Sundararajan; Suresh Sundaram
				Litter detection in Real-World Environments using YOLO models	Espinosa, Alberto; Aguilar-Canto, Fernando; Torres-León, José ; Vergara-Sánchez, Diana Laura; Moreno Armendariz, Marco; Calvo, Hiram
20th March 2025					
Session	Time	Room	#Papers	Title	Authors
oral S1-C	10:00-11:00	Cosmos 3d	5	Reinforcement Learning for Sustainable Maritime Transportation: Reshaping Adaptive Stress Testing	Thomas Steinfeldt Laursen; Ole Mengshoel
long S1-C	11:00-12:00	Cosmos 1&2		Beyond the Smudge: Simulating Opaque and Transparent Automotive Camera Lens Soiling	Tim Eberhardt; Wilhelm Stork; Tim Bruehl; Robin Schwager; Tin Stribor Sohn
				Data-Driven Fraud Detection in Active Participation Programs of End-Use Energy Consumers Based on a Hybrid Fuzzy Approach	Guilherme Carvalho; Zita Vale; Luis Gomes; Ricardo Faia; Edson Costa
				Machine Learning in Sensors: Feature Learning from Complex Spectra in mmWave Radar Signals	Erkan Karakas; Tao Wei; Qing Yang
				QGAPHEnsemble : Combining Hybrid QLSTM Network Ensemble via Adaptive Weighting for Short Term Weather Forecasting	Anuvab Sen; Udayon Sen; Mayukhi Paul; Apurba Prasad Padhy; Sujith Sai; Aakash Mallik; Chhandak Mallick
poster S1	12:00-13:00	Cosmos 1&2	7	Leveraging computational intelligence for optimising multimodal transport systems: the KEYSTONE project	Petrakou, Zoi; Renzi, Giulia; Garcia-Perez, Alexeis; Ramiah, Preetha; Papacharalampous, Alexandros
				Improvement of Wind Power Prediction Using Machine Learning Models	Abbas, Qamar; Ali, Hafiz Muhammad
				Optimal Orientation of Residential Buildings considering Solar Radiation and Wind Direction	Farhadi, Payam
				Probabilistic Allocation of Fast Charging Satations Regarding Multivariate Modeling	Farhadi, Payam
				Optimum PV Array Reconfiguration for Maximum Power Harvest in Partially Shaded Conditions	Koukanmehr, Mahdi; Moghaddas Tafreshi , Seyed Masoud
				Computer Vision-Based Analysis of Interfacial Tensions in Microfluidic Three-Phase Systems	Hatiboglu, Anil; Saiti, Evdokia; Oye, Gisle; Shardt, Nadia
				Search of Chemical Compounds for Organic Thin-film Solar Cells using LLM	Handa, Hisashi*; Vasilevich, Aleksandr; Kuzumoto, Yu; Kihara, Taichi
oral S1-D	14:00-15:00	Cosmos 3c	5	Optimized Control of Single-Stage Grid-Connected PV Inverters with Embedded ANN based MPPT	Yacine Triki; Ali BECHOUCHE; Hamid Seddiki; Djaffar OULD ABDESLAM
long S1-D	15:00-16:00	Cosmos 1&2		Efficient Curation of Invertebrate Image Datasets Using Feature Embeddings and Automatic Size Comparison	Mikko Impiö; Philipp Rehsten; Jenni Raitoharju
				HotSD-Net: Convolutional Neural Network for Estimation of Hot Spots in Photovoltaic Modules via Infrared Imaging	Luis Montanez; Juan Reyes-Luévano; Luis Valentin-Coronado; Daniela Moctezuma; Fernando Lezama; Zita Vale
				Automated Phytosensing: Ozone Exposure Classification Based on Plant Electrical Signals	Till Aust; Eduard Buss; Felix Mohr; Heiko Hamann
				Optics-informed deep learning training for classifying gas absorption spectra on the edge	Manos Kirtas; Loukia Avramelou; Carlos Alonso Ramos; Sara Toxqui Rodriguez; Aitor Villafranca Velasco; Nikolaos Passalis; Christos Pappas; Nikolaos Pleros; ANASTASIOS TEFAS